



CUBAN NATIONAL CHEMISTRY CONTEST

“To work in the sterile when one can do the useful; to deal with the easy when one has spirit to try the difficult, is to strip the dignity from the talent.”

José Martí

Name(s): _____ Last name(s): _____
School: _____
Municipality: _____ Province: _____
CODE: _____ Test: **DAY 2. TENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 10th grade. The exam has 4 problems (problem 4, 6 items; problem 5, 2 items; problem 6, 4 items; problem 7, 7 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

																	13 3A		14 4A	15 5A	16 6A	17 7A	18 8A	
1 H 1.008	2 2A																2 He 4.003							
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18							
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95							
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80							
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3							
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)							
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)							

Problem 4. Thermochemistry I (10 points)

4.1	4.2	4.3	4.4	4.5	4.6	Total	Total
4	4	2	4	4	2	20	10

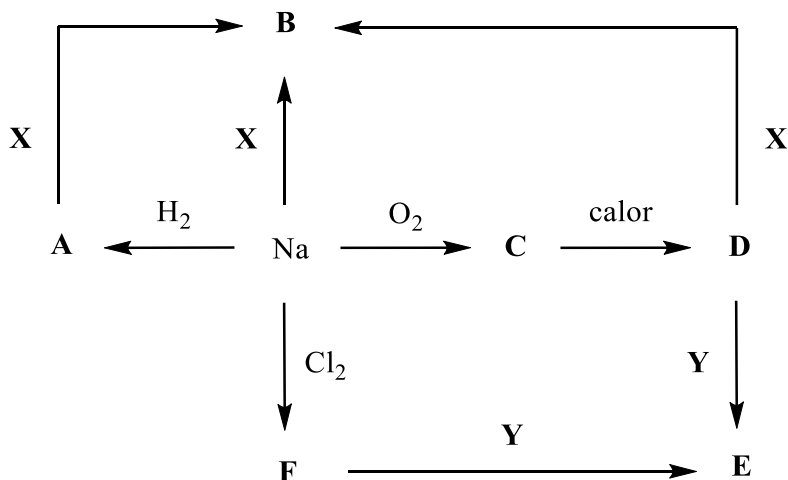
A sample of D-ribose ($\text{C}_5\text{H}_{10}\text{O}_5$, $M = 150,2 \text{ g}\cdot\text{mol}^{-1}$) sugar of mass 0,727 g was placed in a bomb calorimeter and burnt in the presence of excess oxygen. The temperature in the calorimeter increased by 0,910 K. In another experiment using the same calorimeter, the complete combustion of 0,825 g of benzoic acid ($\text{C}_7\text{H}_6\text{O}_2$, $M = 122,1 \text{ g}\cdot\text{mol}^{-1}$) increased the temperature by 1,940 K.

- 4.1. Calculate the heat capacity (in $\text{kJ}\cdot\text{K}^{-1}$) of the calorimeter if the internal energy of combustion of benzoic acid is $-3251 \text{ kJ}\cdot\text{mol}^{-1}$.
- 4.2. Calculate the internal energy of combustion of D-ribose.
- 4.3. Write the chemical equation corresponding to the complete combustion of D-ribose.
- 4.4. Calculate the enthalpy of combustion of D-ribose.
- 4.5. Calculate the enthalpy of formation of D-ribose.
 $\Delta H_f^\circ(\text{H}_2\text{O}(\text{l})) = -286 \text{ kJ}\cdot\text{mol}^{-1}$, $\Delta H_f^\circ(\text{CO}_2(\text{g})) = -394 \text{ kJ}\cdot\text{mol}^{-1}$.
- 4.6. Define the concept of f enthalpy of formation.

Problem 5. Chemical properties (10 points)

5.1	5.2	Total	Total
11	9	20	10

All substances from **A** to **F** that appear in the following scheme contain sodium.



- 5.1. Write the molecular formula of substances **A**, **B**, **C**, **D**, **E**, **F**, **X** and **Y**.
- 5.2. Write the chemical equations corresponding to all the reactions represented in the scheme.

Problem 6. Thermochemistry II (10 points)

6.1	6.2	6.3	6.4	Total	Total
7	5	4	4	20	10

Thermochemical cycles allow us to calculate values of enthalpy variations that occur in processes challenging to carry out experimentally. This is possible because enthalpy is a state function and only depends on the initial and final conditions.

6.1. Use the following data to calculate the lattice energy U of NaCl.



Standard enthalpy of formation:



Standard enthalpy of dissociation:



Electron affinity:



Standard enthalpy of sublimation:



Ionization energy:



In cases where the lattice energy is unknown, it is possible to estimate its value theoretically by employing the Kapustinskii equation.

$$U = \frac{1,214 \cdot 10^5 \cdot z_c \cdot z_a \cdot v}{r_c + r_a} \left(1 - \frac{34,5}{r_c + r_a} \right)$$

Where U is the lattice energy in $\text{kJ} \cdot \text{mol}^{-1}$, z_c y z_a are the cation and anion charges, r_c and r_a are the cation and anion radius in pm, and v is the total number of ions in the empirical formula.

6.2. Calculate the lattice energy of NaCl with the Kapustinskii equation. Which is the physical meaning of the negative term in the equation? ($r(\text{Na}^+) = 102 \text{ pm}$ y $r(\text{Cl}^-) = 181 \text{ pm}$).

6.3. Substances AgCl and NaCl have the same crystalline structure in the solid state. Both are formed by the same anion and have cations with equal charge and similar radius ($r(\text{Ag}^+) = 115 \text{ pm}$). How can you explain the fact that the lattice energies of these compounds are so different? $U(\text{AgCl}) = -912 \text{ kJ} \cdot \text{mol}^{-1}$.

6.4. Compare the solubility in water and the melting point of the substances NaCl and AgCl.

Problem 7. Chemical bonding (20 points)

7.1	7.2	7.3	7.4	7.5	7.6	7.7	Total	Total
6	8	6	4	5	5	6	40	20

The nonmetallic chemical element X forms oxides **A** and **B**. The ratio between the mass percent of X in **A** and the mass percent of X in **B** is equal to 1,25. When the previous oxides react with water, they originate the oxyacids **C** (39,1% in mass of X) and **D** (32,7% in mass of X), respectively, being **D** the most acidic of both.

7.1. Find out which element is X. Justify numerically.

7.2. Write the formula of compounds **A**, **B**, **C** y **D**.

7.3. The high acidity of **D** can be explained due to the stability of the mono negative anion that is formed when losing an H^+ ion. Draw the resonance structures of the mentioned anion.

7.4. Predict the existence and magnetic properties of the hypothetical molecule X_2 . Use the postulates of the Theory of Molecular Orbital.

7.5. Represent the electronic structure of molecule **A** according to the postulates of the Valence Bond Theory and employing hybrid orbitals.

7.6. In addition, element X forms ions $X_2O_3^{2-}$ and $X_2O_8^{2-}$. Draw the Lewis structures of both chemical species.

7.7. When substance XCl_4 reacts with oxygen **E** is form, having the formula $XOCl_2$. Draw the Lewis structure of **E**. Predict the molecular geometry and the bond angle.